

Plant Root Phenotyping Pipeline for Tomato.

ChronoRootAnalysis

Plant Analysis

Analysis Overview

Plant Overlay

Generate Report

Report

Individual plant root analysis

Select Project Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/Tomato

Select Video Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/rpi104_tomato/1

Raspberry Module

104

(the raspberry module identifier)

Camera

1

(the plate/camera identifier)

Plant Number

(a number to identify plants in the plate, left to right)

Identifier

(experiment identifier, e.g. WT, Col0)

Analysis and postprocessing parameters

☐ Save Cropped Images

☐ Video has QR codes

Manual Calibration Parameters:

Open Calibration Helper

Known (mm):

10

Pixels:

240

Set processing limit

0

(in days, 0 means no limit)

Capture interval

15

(in minutes, usually 15 minutes)

Emergence distance

2

(in millimeters, default: 2 mm)

Save

Preview video

Analyze Plant

Process all plants

Load config json from file

Load previous configuration

Launch the interface, this will open the main analysis window.

Interface Components:

- **Select Project Folder:** Specifies the storage location for your analysis results. A project encompasses a complete experimental setup involving one or more Raspberry Pi modules.
- **Select Video Folder:** Identifies the video file for current processing. Videos must be processed sequentially, one at a time.
- **Plant Identification Fields:** The Raspberry Pi Module, Camera, Plant Number, and Identifier fields allow you to tag each plant according to its origin (camera and module), position on the plate, and biological variety.
- **Manual Calibration Parameters:** If your video does not have a 1-cm QR code, measure a known distance and write the pixels from the calibration helper.
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Getting Started:

Begin by creating a dedicated folder for storing results, then select the first video from the Demo dataset. A popup window will prompt you to choose the appropriate folders.

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Select Project Folder

/Demo/rpi104_tomato

Select Video Folder

/Demo/rpi104_tomato/1

Raspberry Module

104

(should be the raspberry number)

Camera

1

(should be the camera number)

Plant Number

1

(should be a number, to identify plant)

Identifier

A

(variety identifier, e.g. WT, Col0)

Save

Preview video

Analyze Plant

Process all plants

Load config json from file

Load previous configuration

Analysis and postprocessing parameters

☐ Save Cropped Images

☐ Video has QR codes

Set processing limit

0

(in days, 0 means no limit)

Capture interval

15

(in minutes, usually 15 minutes)

Emergence distance

2

(in millimeters, default: 2 mm)

Manual Calibration Parameters:

Open Calibration Helper

Known (mm):

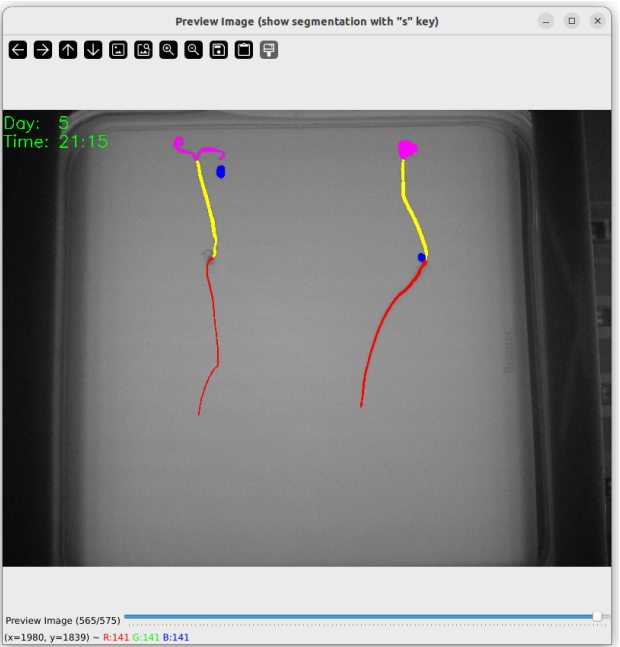
10

Pixels:

240

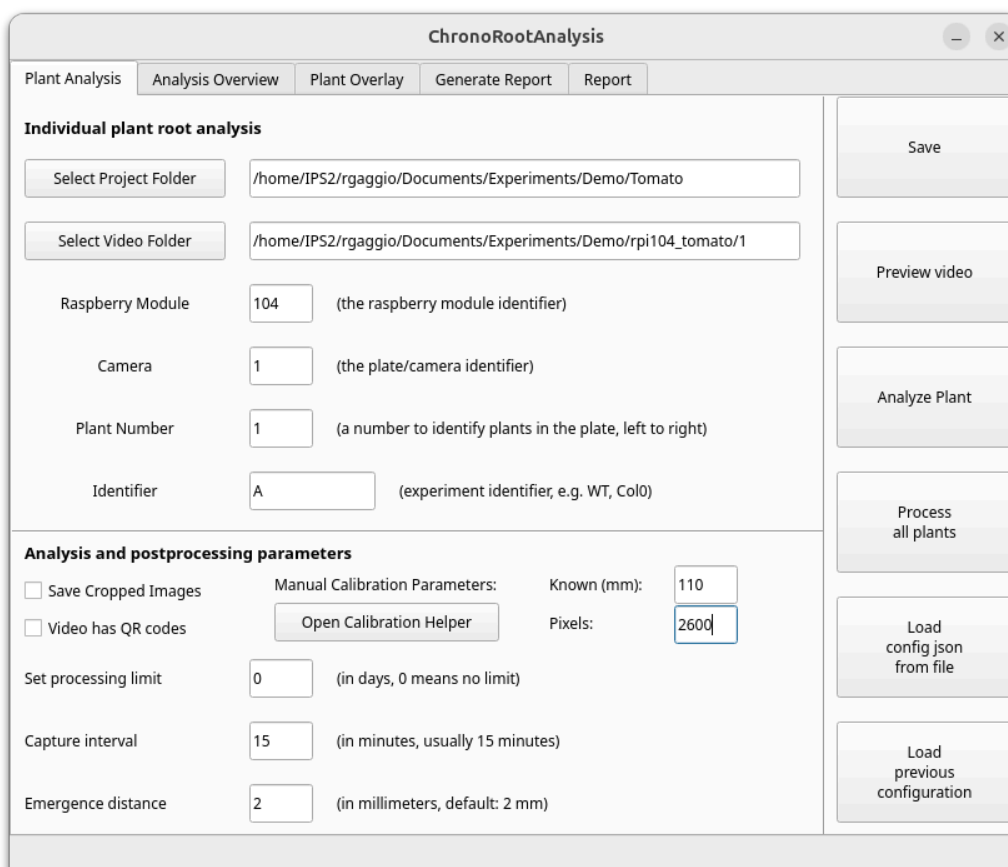
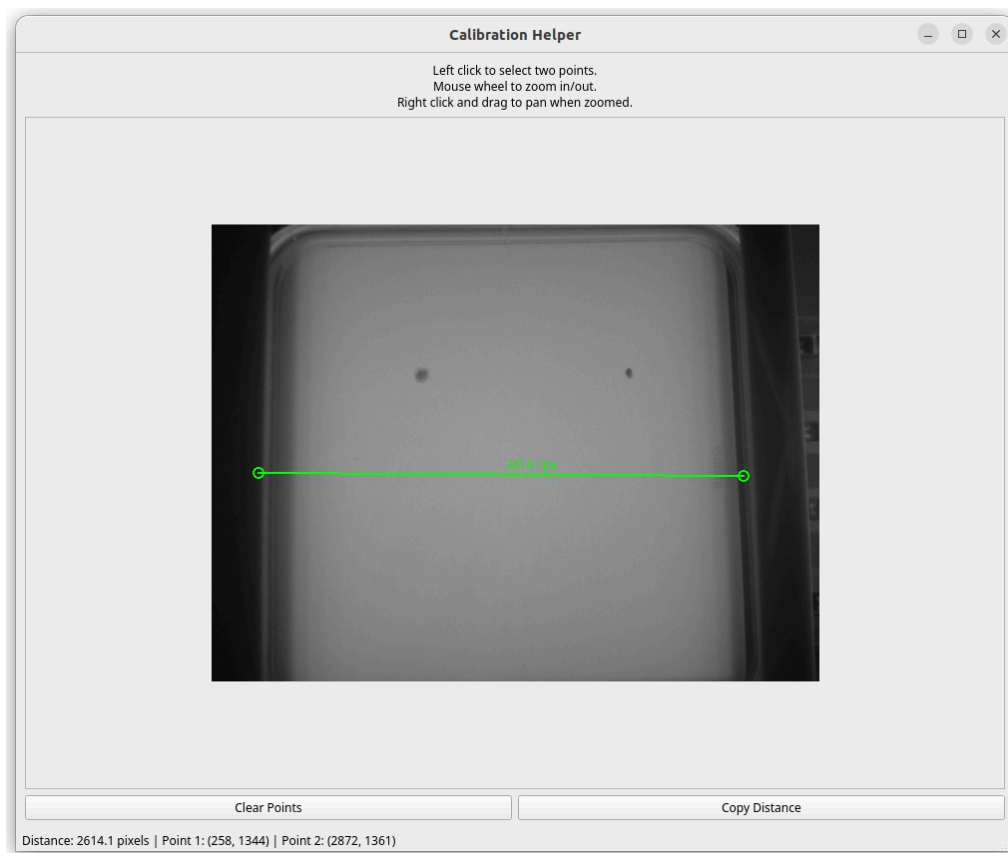
Video Preview and Setup:

Click the "Preview Video" button to view your selected video. Use the scrollbar to navigate between frames and press the "S" button to toggle the segmentation overlay for visual inspection. In this case, we can see that the video contains a QR code.



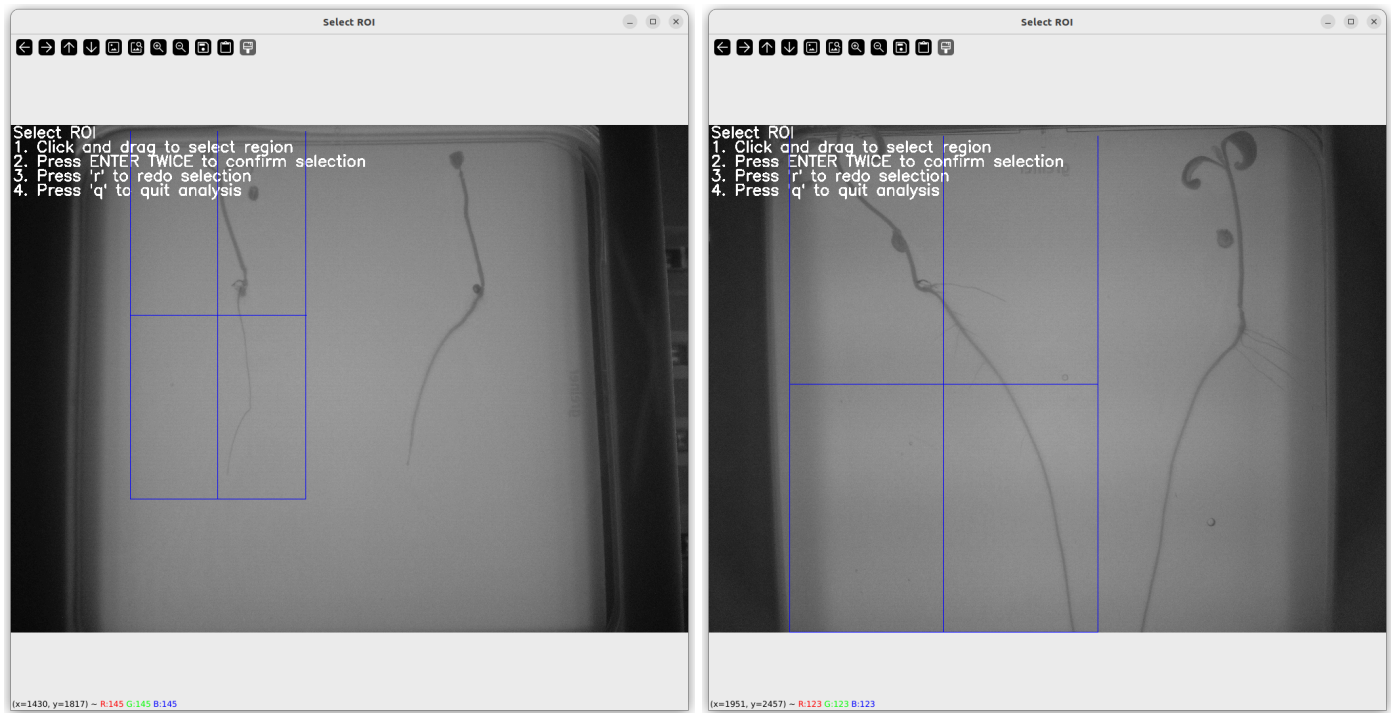
Using the calibration helper to complete the distances:

As these demo plates do not contain QR codes we can use the known distance on the center of the plate, the visible area given by the plate holder is 11 centimeters, and according to the calibrator, those measure ~2600 pixels.

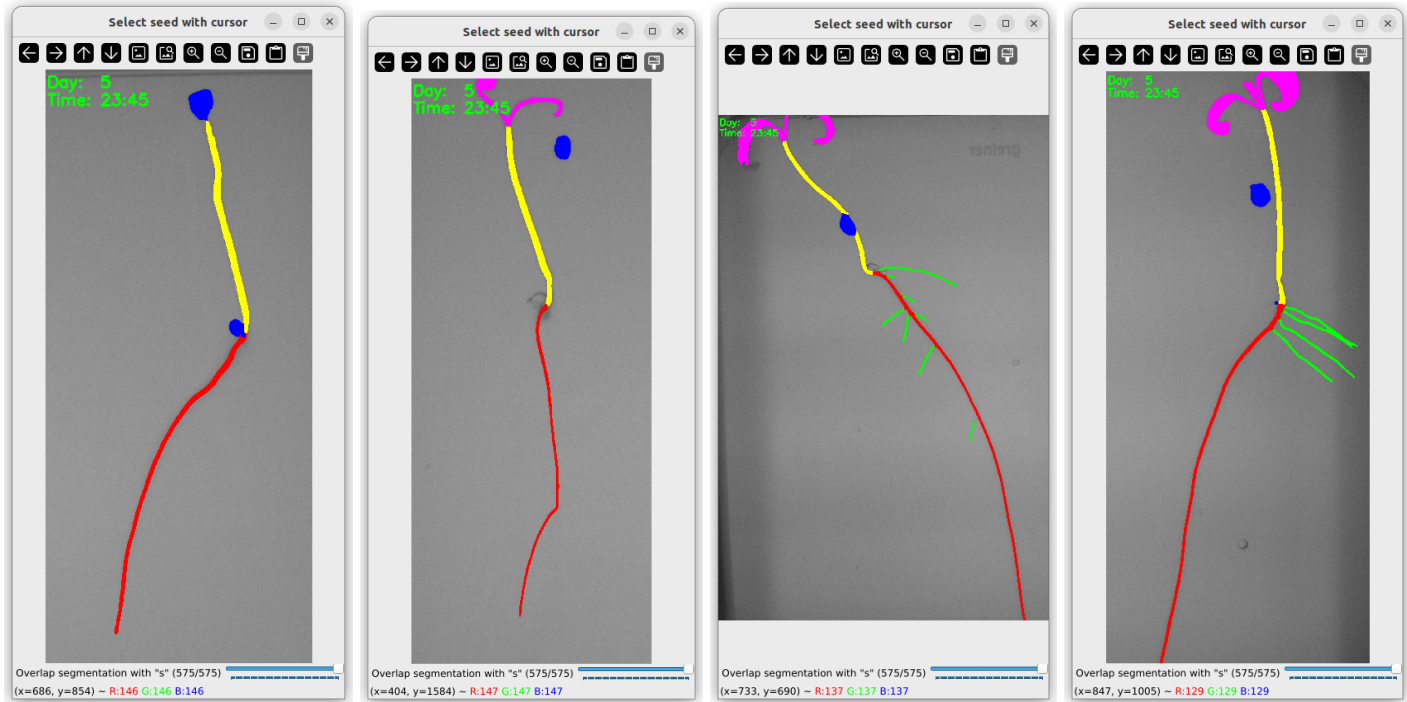


Plant Analysis Workflow:

1. ROI Selection: Click "Analyze Plant" to open the ROI (Region of Interest) selector. Manually define the analysis area and press double-enter to confirm your selection.

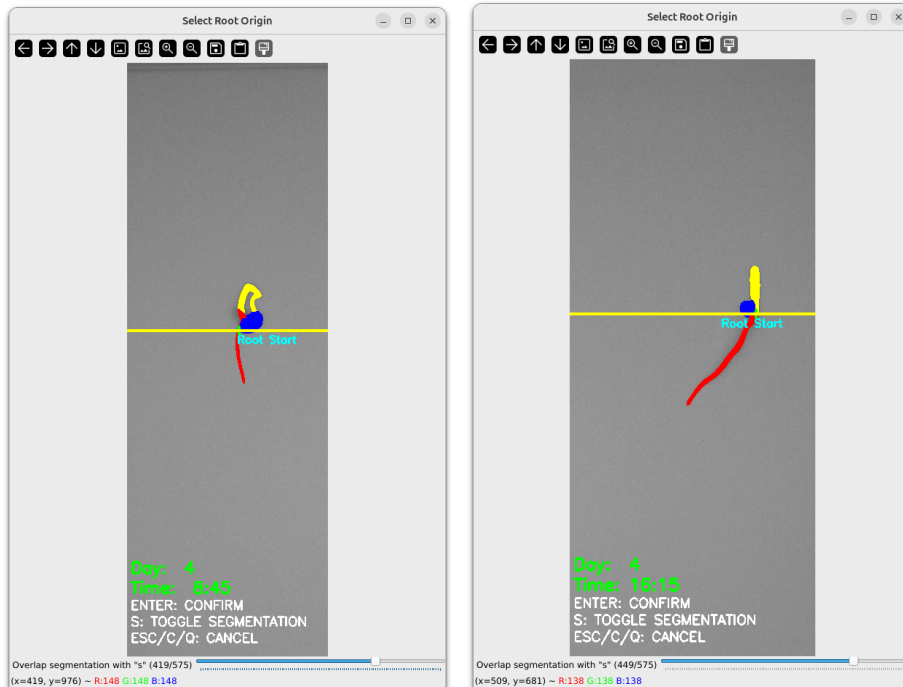


2. Evaluate the segmentation quality: Use this step to determine the correct amount of days that the segmentation can be properly analyzed. In some cases here we can determine any potential issues. In this case we can see that we need to be careful with the main root positioning as we may era some potential lateral roots.

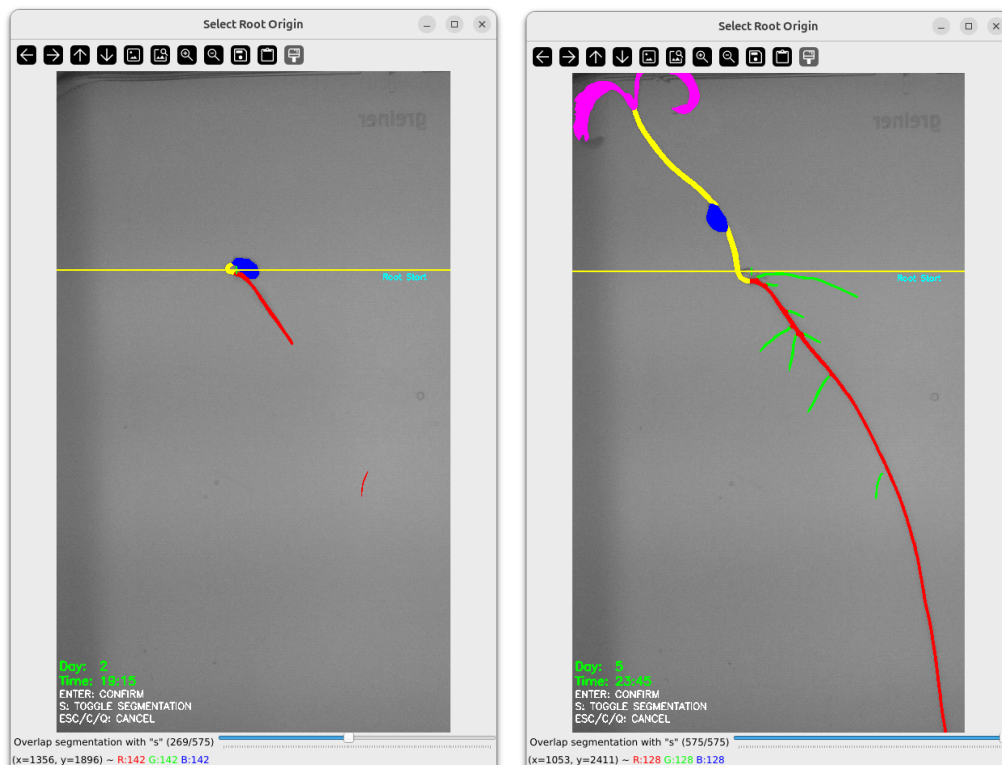


3. Select the point of main root start: This acts as a cleaning step, everything above the selected point will be discarded and not regarded as part of the root system. Please note that you can zoom in using the mouse wheel.

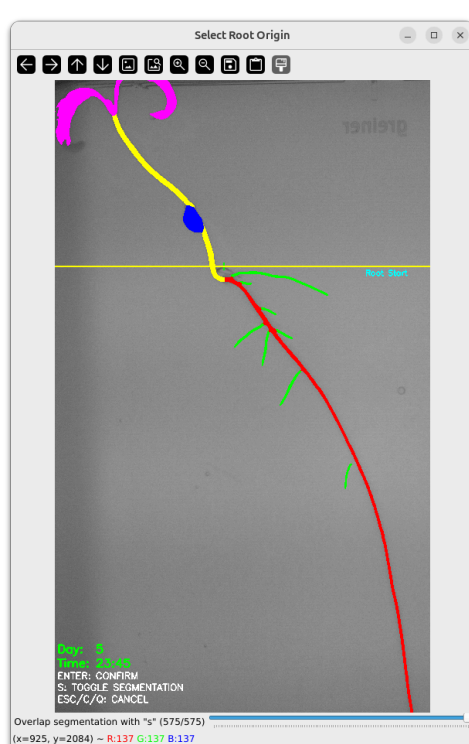
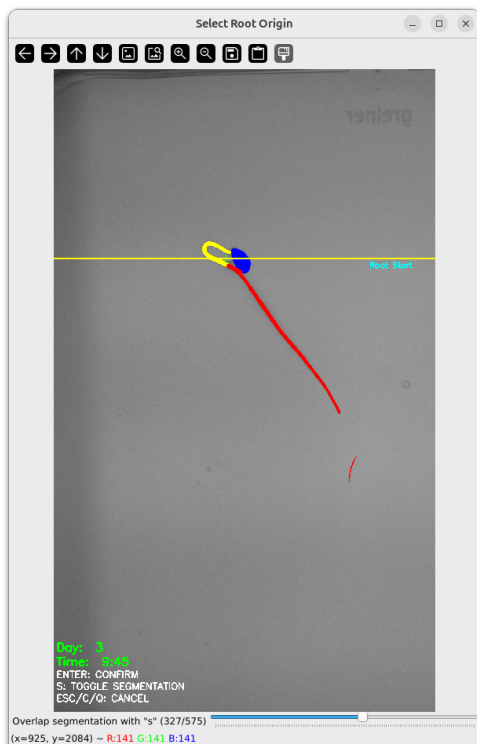
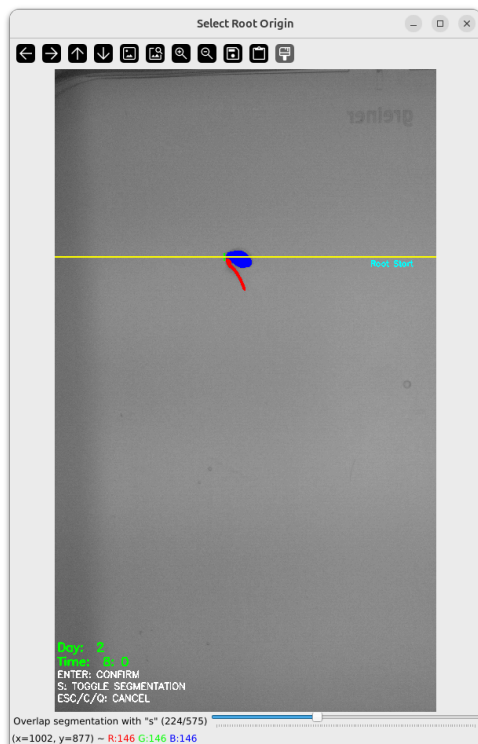
We select the two plants of the first camera with identifier A.



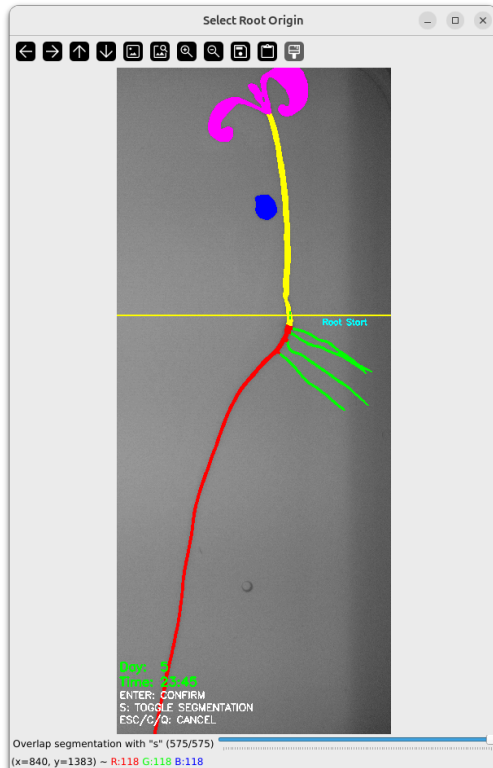
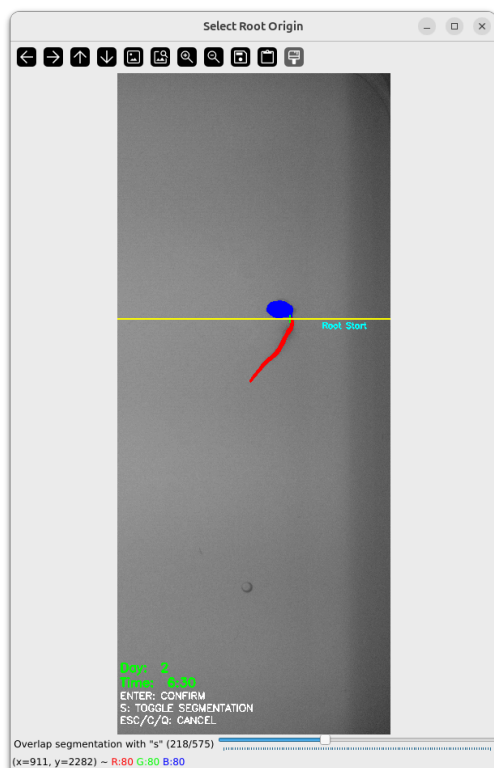
For the second camera, root start selection can be trickier.



But in this kind of cases, do not worry! Selecting above the root will also help, as we can see there are not any miss-detections that can cause us any trouble. The erasure of the above part can be



Then the last plant:



Quality Control and Batch Processing:

Navigate to the "Analysis Overview" tab to monitor experiment progress and error rates. Poor ROI selection or incorrect seed point positioning may cause errors in graph creation. Once individual plant processing is complete, select "Process all plants."

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Analysis Overview

Plant Overlay

Generate Report

Report

	Experiment	Raspberry	Camera	Plant Number	Result ID	Error Rate	Status	Finish Date
1	A	104	cam_1	plant_1	Results_0	0.0	Finished	2026-01-13 18:15:03...
2	A	104	cam_1	plant_2	Results_0	0.0	Finished	2026-01-13 18:15:56...
3	B	104	cam_2	plant_1	Results_0		Not finished	
4	B	104	cam_2	plant_2	Results_0		Not finished	

Refresh

Open Path

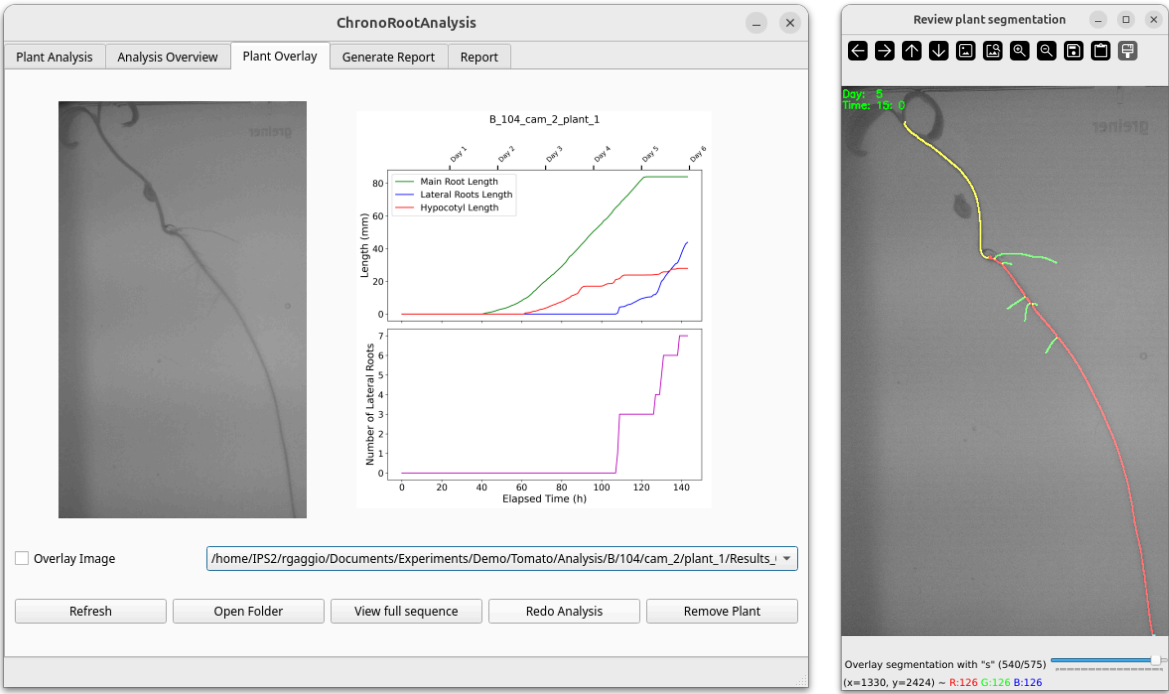
Remove Plant

Repeat Analysis

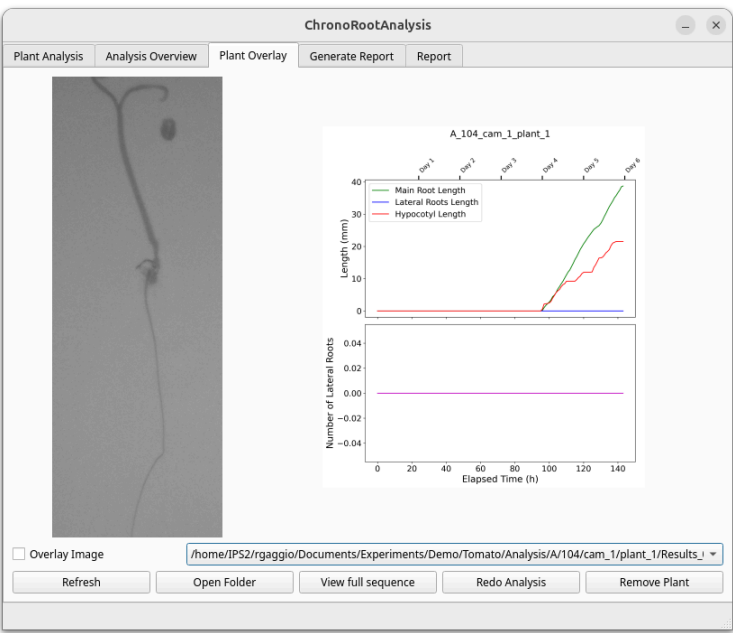
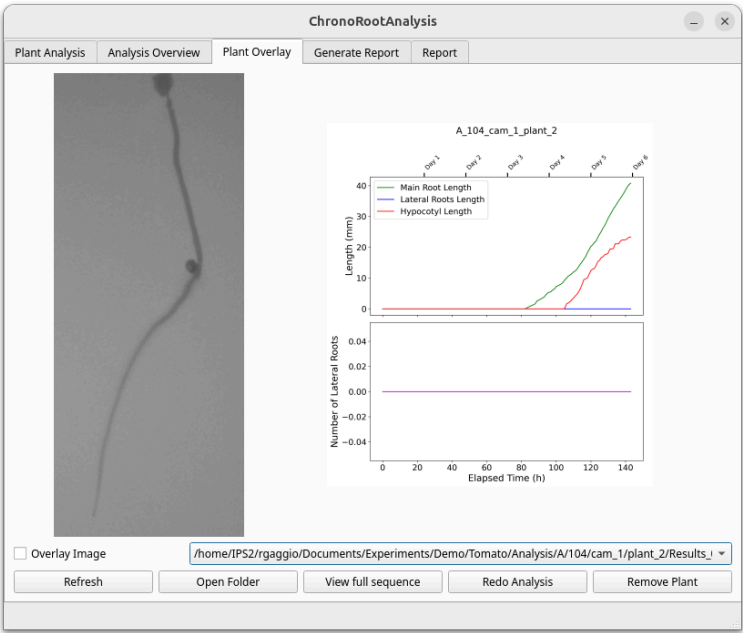
Process all plants

Visual Inspection:

The "Plant Overlay" tab provides visual inspection capabilities for both segmentation results and plant root performance analysis. This interface allows you to discard problematic plants or restart the entire process if needed.



Above we can see a stop of growth in the main root due to the plant reaching the bottom of the plate. This should normally mark the end of the experiment. Meanwhile, plants of plate A still have not reached the base; we can see this as the main root length does not stagnate. But we need to be careful, because that can also happen if the bounding box is not properly chosen (does not give space to grow).



Report Generation:

After completing individual plant analyses, you can generate comprehensive reports. The demo dataset includes 5 complete days of plant growth data—feel free to experiment with different report types. The system can perform interval testing (for example, every 6 hours) to compare different varieties using statistical methods such as the Mann-Whitney U test to determine if varieties show statistically significant differences.

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Report

Select Project Folder

/home/IPS2/rgaggio/Documents/Experiments/Demo/Tomato

☒ **Average intervals before testing**

Time series stats interval (dt, in hours)

6

Hypothesis testing uses Mann-Whitney test every dt interval. If selected, an average value will be used, or a step (i*dt) otherwise

☒ **Perform Functional PCA on time series**

☐ Normalize FPCA Boxplots

Number of components

2

☒ **Do Convex hull analysis**

☐ Save images for each day

Days to report

3,4,5

(Numbers separated by commas)

☐ **Evaluate Growth Speeds and perform Fourier Analysis**

Speeds stats interval (dt, in hours)

6

☐ **Do Lateral Root Angles Analysis**

Emergence distance

2

(in millimeters, default: 2 mm)

Days to report

First LR Tip Stats interval (dt, in hours)

6

Processing limit

0

Capture interval

15

Save

Process all plants

Generate report

Load previous configuration

The system automatically generates comprehensive reports containing various figure types. All raw figures and images are saved within the report directory for detailed analysis and further investigation.

